3.5.2 Contextual factors

The following sections describe the three contextual authentication factors in detail, outlining their typical uses, strengths, and limitations.

BEHAVIOR FACTOR—SOMETHING YOU DO, LIKE MOUSE MOVEMENTS

Behavior factors analyze patterns of a person's behavior rather than static credentials. These patterns include typing speed, keystroke dynamics, mouse movement, and gait recognition. You may have been challenged by a CAPTCHA asking you to click a box and thought, "How is this proving I'm a human?" These challenges can distinguish a human's mouse movements from a machine's—at least for now. They are a good example of a widely deployed behavior factor.

LOCATION FACTOR—SOMEWHERE YOU ARE LOCATED, LIKE GEOLOCATION

Location factors use things like GPS, IP addresses, or Wi-Fi locations to verify requests. They can prevent access from unexpected locations or enforce geo-restrictions but are vulnerable to VPNs and GPS spoofing. Location factors are most effective when used in combination with one or more primary factors. One application of location factors is conducting velocity checks to detect impossible travel, where requests originate from locations that are too distant for travel between them.

TEMPORAL FACTOR—THE TIME YOU'RE IN, SUCH AS THE CURRENT LOCAL TIME

Temporal factors assess whether an authentication request occurs at an appropriate time. They can identify logins at unusual hours or detect anomalies in session timing. Like other contextual factors, they are most effective when combined with other primary factors. AI-based systems have made it easier to identify anomalous access times.